



DC – AI – Routing – Switching – Security – Wireless – Automation

PTX12008 Power Distribution and Optimization

An HPE Juniper Networking Techpost

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HPE



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1 PTX12008 Power Distribution and Optimization

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Learn how the PTX12008 optimizes power consumption through automatic and operator-controlled mechanisms such as adaptive fan management, unused port shutdown, selective PFE offlining, MACsec clock gating, and dynamic fabric or line card power control to improve energy efficiency in high-capacity networks.

1.1 Introduction

The PTX12008 is a 8-slot modular chassis running Junos-EVO. It inherits the of power optimization/efficiency features introduced and validated on the Junos-EVO.

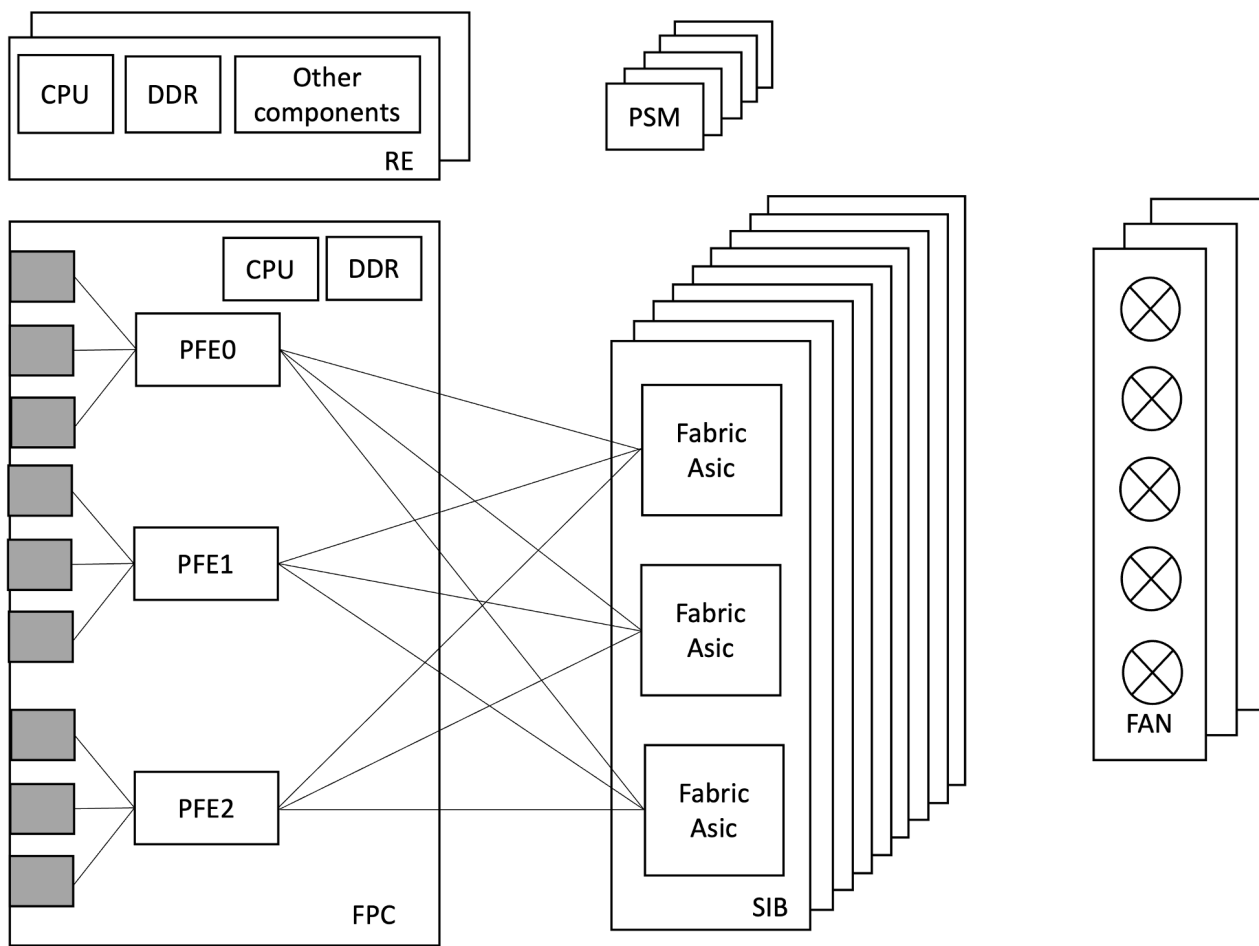
Power efficiency is increasingly important as network capacity is growing. The EVO platform integrates granular power control at every level of the forwarding pipeline from individual port SerDes to entire line cards so operators can match power draw to actual traffic demand without sacrificing availability.

Key principles:

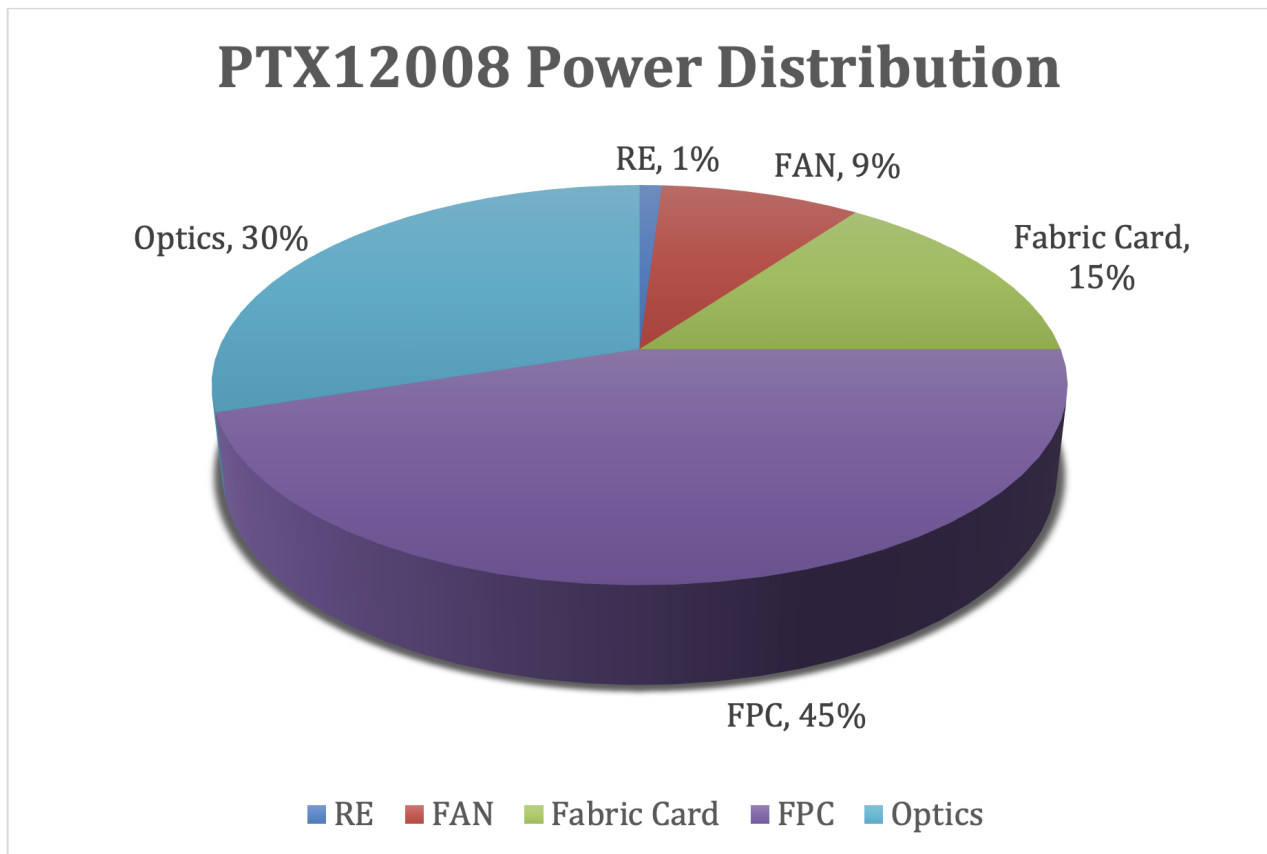
1. Many optimizations are enabled by default and require no configuration.
2. Operator-driven optimizations allow further tuning for specific deployments.
3. PFE-level granularity in optimizations helps minimal blast radius.
4. All power-off operations on FRUs i.e. Fabric card/Line card/PFE can be applied/reversed without system reboot.

1.2 Power Consumers

The PTX12008 is a modular system. At high level, the major power consumers are Line cards, Optics, Fabric Cards, Cooling Fans and Routing Engines.



Here is power distribution across these components assuming each optics consumes 30W power (Typical 800G ZR optics power consumption)



Factors that increase power consumption:

- Higher ambient temperature (fans speed up to cool down system)
- Higher elevation
- More number of optics inserted
- High power optics i.e. ZR HP
- Higher traffic load in bps and pps
- Using MACsec on Wan Ports
- Fully populated FPC slots

Factors that decrease power consumption:

- Empty port slots (SerDes auto-shutdown)
- Ports configured “unused”
- PFEs powered off when not needed
- Fabric card/Line card offline in empty or temporarily unused slots
- Low traffic periods (PFE and fabric SerDes run at lower utilization)

1.3 Power Saving Features in PTX12008

1.4 Environment Management Policy - Controlling FAN speed to save Power

The Environmental Monitoring (EM) Policy is a built-in subsystem that continuously monitors thermal sensors distributed throughout the PTX12008 chassis and dynamically adjusts fan speed in real time. Because Fans are significant power consumers running them only as fast as thermals require is one of the most effective automatic power saving mechanisms in the platform.

How EM Policy Works:

1. Thermal sensors are placed at key locations inside the chassis: Line card inlet/exhaust, Fabric card, RE, CB, PSM, and fan tray zones.
2. The EM Policy daemon collects sensor readings at regular intervals and computes the required airflow.
3. Fan speed is set to the minimum level that keeps all components within their thermal operating limits.
4. As temperature rises due to higher ambient, more traffic load or more optics active, the EM Policy incrementally increases fan speed.
5. As temperature falls due to lower ambient, PFEs/optics powered off or traffic drops, fan speed is reduced automatically to save power.
6. This feedback loop runs continuously without any operator intervention.

As per EM Policy, FANs runs at minimum speed required to keep system thermal within limits helps to save significant power. Here is an example that shows the difference in power consumption when the Fan runs at ~60% vs. 100%.

```

1 root@ptx12008> show chassis fan
2   Item Status % RPM Measurement
3   Fan Tray 0 Fan 0 OK 56% 5700 RPM
4   Fan Tray 0 Fan 1 OK 62% 7200 RPM
5   Fan Tray 0 Fan 2 OK 55% 5550 RPM
6   Fan Tray 0 Fan 3 OK 62% 7200 RPM
7   Fan Tray 0 Fan 4 OK 55% 5550 RPM
8   Fan Tray 0 Fan 5 OK 63% 7350 RPM
9   Fan Tray 0 Fan 6 OK 55% 5550 RPM
10  Fan Tray 0 Fan 7 OK 62% 7200 RPM
11  Fan Tray 0 Fan 8 OK 55% 5550 RPM
12  Fan Tray 0 Fan 9 OK 62% 7200 RPM
13  Fan Tray 1 Fan 0 OK 56% 5700 RPM
14  Fan Tray 1 Fan 1 OK 62% 7200 RPM
15  Fan Tray 1 Fan 2 OK 56% 5700 RPM
16  Fan Tray 1 Fan 3 OK 62% 7200 RPM
17  Fan Tray 1 Fan 4 OK 56% 5700 RPM
18  Fan Tray 1 Fan 5 OK 62% 7200 RPM
19  Fan Tray 1 Fan 6 OK 55% 5550 RPM
20  Fan Tray 1 Fan 7 OK 62% 7200 RPM
21  Fan Tray 1 Fan 8 OK 56% 5700 RPM
22  Fan Tray 1 Fan 9 OK 61% 7050 RPM
23  Fan Tray 2 Fan 0 OK 56% 5700 RPM
24  Fan Tray 2 Fan 1 OK 62% 7200 RPM
25  Fan Tray 2 Fan 2 OK 55% 5550 RPM
26  Fan Tray 2 Fan 3 OK 62% 7200 RPM
27  Fan Tray 2 Fan 4 OK 56% 5700 RPM
28  Fan Tray 2 Fan 5 OK 61% 7050 RPM
29  Fan Tray 2 Fan 6 OK 55% 5550 RPM
30  Fan Tray 2 Fan 7 OK 62% 7200 RPM
31  Fan Tray 2 Fan 8 OK 56% 5700 RPM
32  Fan Tray 2 Fan 9 OK 62% 7200 RPM
33
34 root@ptx12008> show chassis power detail | match FAN
35   Fan Tray 0 269
36   Fan Tray 1 254
37   Fan Tray 2 265

```

When Fan runs at ~60%, Fan trays consume a total of ~788 watts.

```

1 root@ptx12008> show chassis fan
2   Item Status % RPM Measurement
3   Fan Tray 0 Fan 0 OK 100% 10200 RPM
4   Fan Tray 0 Fan 1 OK 100% 11700 RPM
5   Fan Tray 0 Fan 2 OK 100% 10200 RPM
6   Fan Tray 0 Fan 3 OK 100% 11700 RPM
7   Fan Tray 0 Fan 4 OK 100% 10200 RPM
8   Fan Tray 0 Fan 5 OK 100% 11700 RPM
9   Fan Tray 0 Fan 6 OK 100% 10050 RPM
10  Fan Tray 0 Fan 7 OK 100% 11700 RPM
11  Fan Tray 0 Fan 8 OK 100% 10050 RPM
12  Fan Tray 0 Fan 9 OK 100% 11700 RPM
13  Fan Tray 1 Fan 0 OK 100% 10200 RPM
14  Fan Tray 1 Fan 1 OK 100% 11700 RPM

```

```

15 Fan Tray 1 Fan 2 OK 100% 10200 RPM
16 Fan Tray 1 Fan 3 OK 100% 11700 RPM
17 Fan Tray 1 Fan 4 OK 100% 10050 RPM
18 Fan Tray 1 Fan 5 OK 100% 11700 RPM
19 Fan Tray 1 Fan 6 OK 100% 10200 RPM
20 Fan Tray 1 Fan 7 OK 100% 11550 RPM
21 Fan Tray 1 Fan 8 OK 100% 10050 RPM
22 Fan Tray 1 Fan 9 OK 100% 11550 RPM
23 Fan Tray 2 Fan 0 OK 100% 10050 RPM
24 Fan Tray 2 Fan 1 OK 100% 11700 RPM
25 Fan Tray 2 Fan 2 OK 100% 10200 RPM
26 Fan Tray 2 Fan 3 OK 100% 11700 RPM
27 Fan Tray 2 Fan 4 OK 100% 10200 RPM
28 Fan Tray 2 Fan 5 OK 100% 11700 RPM
29 Fan Tray 2 Fan 6 OK 100% 10200 RPM
30 Fan Tray 2 Fan 7 OK 100% 11550 RPM
31 Fan Tray 2 Fan 8 OK 100% 10200 RPM
32 Fan Tray 2 Fan 9 OK 100% 11550 RPM
33
34 root@ptx12008> show chassis power detail | match FAN
35 Fan Tray 0 1363
36 Fan Tray 1 1343
37 Fan Tray 2 1375

```

When Fan runs at ~100%, Fan trays consume a total of ~4,081watts.

This example shows how running Fan as least as required helps to save system power significantly.

1.5 Wan Ports “unused” Configuration

A port populated with a transceiver but not in service can be disabled via the “unused” keyword. This shuts down the forwarding path of the transceiver and help to save power. This “unused” configuration removes the interface from all “show interfaces” output, but optic module remains powered for inventory/monitoring purposes.

Here is an example that shows the power saving by marking ports “unused”.

```

1 root@ptx12008> show chassis hardware
2 Hardware inventory:
3 Item Version Part number Description
4 Chassis EX352 PTX12008
5 Midplane 0 REV 05 750-127042 BCCG6098 Unsupported
6 PSM 2 Rev 01 740-129297 1F31B290037 3000W AC AFI Power Supply
7 PSM 3 Rev 01 740-129297 1F31B330071 3000W AC AFI Power Supply
8 PSM 12 Rev 01 740-129297 1F31B290047 3000W AC AFI Power Supply
9 PSM 13 REV 01 740-129297 1F31B480109 3000W AC AFI Power Supply
10 PSM 15 REV 01 740-129297 1F31B480061 3000W AC AFI Power Supply
11 PSM 16 Rev 01 740-129297 1F31B330010 3000W AC AFI Power Supply
12 PSM 17 Rev 01 740-129297 1F31B330167 3000W AC AFI Power Supply
13 PSM 18 Rev 01 740-129297 1F31B330106 3000W AC AFI Power Supply
14 Routing Engine 0 BUILTIN BUILTIN PTX12K-RE
15 Routing Engine 1 BUILTIN BUILTIN PTX12K-RE
16 CB 0 REV 06 750-173945 BCFS1409 Control Board
17 CB 1 REV 06 750-173945 BCFS1414 Control Board
18 FPC 0 REV 16 750-173325 BCFY9869 PTX12K-LC54QDD
19 CPU REV 04 750-173947 BCFW4413 PTX12K-LC54 PMB Board
20 PIC 0 BUILTIN BUILTIN PTX12K-54QDD800-LC-PIC
21 Xcvr 0 REV 21 750-152172 ZP3924440660 QSFP-DD800-ACTIVE-E-LPBK
22 Xcvr 1 REV 21 750-152172 ZP3924440684 QSFP-DD800-ACTIVE-E-LPBK
23 Xcvr 2 REV 21 750-152172 ZP3924380165 QSFP-DD800-ACTIVE-E-LPBK
24 Xcvr 20 REV 21 750-152172 ZP3924440662 QSFP-DD800-ACTIVE-E-LPBK
25 Xcvr 21 REV 21 750-152172 ZP3924440689 QSFP-DD800-ACTIVE-E-LPBK
26 Xcvr 35 REV 21 750-152172 ZP3924380226 QSFP-DD800-ACTIVE-E-LPBK
27 Xcvr 52 REV 21 750-152172 ZP3924380225 QSFP-DD800-ACTIVE-E-LPBK
28 Xcvr 53 REV 21 750-152172 ZP3924440363 QSFP-DD800-ACTIVE-E-LPBK
29 MEZZ 0 REV 04 750-175158 BCFV8081 PTX12K-LC54QDD MEZZ A Board
30 MEZZ 1 REV 03 750-175159 BCFV9118 PTX12K-LC54QDD MEZZ B Board
31 MEZZ 2 REV 04 750-175033 BCFW6350 PTX12K-LC54 Power Board Right
32 MEZZ 3 REV 05 750-175031 BCFY9285 PTX12K-LC54 Power Board Left
33 SIB 0 REV 09 750-173308 BCER8121 SIB-PTX12008-SF
34 MEZZ 0 REV 03 750-176911 BCFZ3105 SIB-PTX12008-SF Power Board
35 Fan Tray 0 REV 13 750-127041 BCCM0708 PTX12008 Fan tray

```

```

36 Fan Tray 1 REV 08 750-127041 BCCG8484 PTX12008 Fan tray
37 Fan Tray 2 REV 08 750-127041 BCCG8481 PTX12008 Fan tray
38 POWERSHELF 0 BUILTIN BUILTIN 5 Row PSM POWERSHELF
39
40 root@ptx12008> show interfaces terse et-0/0/0
41 Interface Admin Link Proto Local Remote
42 et-0/0/0 up up
43 et-0/0/0.16386 up up multiservice
44
45 root@ptx12008> show interfaces terse et-0/0/1
46 Interface Admin Link Proto Local Remote
47 et-0/0/1 up up
48 et-0/0/1.16386 up up multiservice
49
50 {master}
51 root@ptx12008> show interfaces terse et-0/0/2
52 Interface Admin Link Proto Local Remote
53 et-0/0/2 up up
54 et-0/0/2.16386 up up multiservice
55
56 root@ptx12008> show chassis power detail | match FPC
57 FPC 0 1234
58
59 root@ptx12008# set interfaces et-0/0/0 unused
60
61 root@ptx12008# set interfaces et-0/0/1 unused
62
63 root@ptx12008# set interfaces et-0/0/2 unused
64
65 root@ptx12008# commit
66 re0:
67 configuration check succeeds
68 re1:
69 configuration check succeeds
70 commit complete
71 re0:
72 commit complete
73
74 root@ptx12008> show chassis hardware
75 Hardware inventory:
76 Item Version Part number Serial number Description
77 Chassis EX352 PTX12008
78 Midplane 0 REV 05 750-127042 BCCG6098 Unsupported
79 PSM 2 Rev 01 740-129297 1F31B290037 3000W AC AFI Power Supply
80 PSM 3 Rev 01 740-129297 1F31B330071 3000W AC AFI Power Supply
81 PSM 12 Rev 01 740-129297 1F31B290047 3000W AC AFI Power Supply
82 PSM 13 REV 01 740-129297 1F31B480109 3000W AC AFI Power Supply
83 PSM 15 REV 01 740-129297 1F31B480061 3000W AC AFI Power Supply
84 PSM 16 Rev 01 740-129297 1F31B330010 3000W AC AFI Power Supply
85 PSM 17 Rev 01 740-129297 1F31B330167 3000W AC AFI Power Supply
86 PSM 18 Rev 01 740-129297 1F31B330106 3000W AC AFI Power Supply
87 Routing Engine 0 BUILTIN BUILTIN PTX12K-RE
88 Routing Engine 1 BUILTIN BUILTIN PTX12K-RE
89 CB 0 REV 06 750-173945 BCFS1409 Control Board
90 CB 1 REV 06 750-173945 BCFS1414 Control Board
91 FPC 0 REV 16 750-173325 BCFY9869 PTX12K-LC54QDD
92 CPU REV 04 750-173947 BCFW4413 PTX12K-LC54 PMB Board
93 PIC 0 BUILTIN BUILTIN PTX12K-54QDD800-LC-PIC
94 Xcvr 0 REV 21 750-152172 ZP3924440660 QSFP-DD800-ACTIVE-E-LPBK
95 Xcvr 1 REV 21 750-152172 ZP3924440684 QSFP-DD800-ACTIVE-E-LPBK
96 Xcvr 2 REV 21 750-152172 ZP3924380165 QSFP-DD800-ACTIVE-E-LPBK
97 Xcvr 20 REV 21 750-152172 ZP3924440662 QSFP-DD800-ACTIVE-E-LPBK
98 Xcvr 21 REV 21 750-152172 ZP3924440689 QSFP-DD800-ACTIVE-E-LPBK
99 Xcvr 35 REV 21 750-152172 ZP3924380226 QSFP-DD800-ACTIVE-E-LPBK
100 Xcvr 52 REV 21 750-152172 ZP3924380225 QSFP-DD800-ACTIVE-E-LPBK
101 Xcvr 53 REV 21 750-152172 ZP3924440363 QSFP-DD800-ACTIVE-E-LPBK
102 MEZZ 0 REV 04 750-175158 BCFV8081 PTX12K-LC54QDD MEZZ A Board
103 MEZZ 1 REV 03 750-175159 BCFV9118 PTX12K-LC54QDD MEZZ B Board
104 MEZZ 2 REV 04 750-175033 BCFW6350 PTX12K-LC54 Power Board Right
105 MEZZ 3 REV 05 750-175031 BCFY9285 PTX12K-LC54 Power Board Left
106 SIB 0 REV 09 750-173308 BCER8121 SIB-PTX12008-SF
107 MEZZ 0 REV 03 750-176911 BCFZ3105 SIB-PTX12008-SF Power Board
108 Fan Tray 0 REV 13 750-127041 BCCM0708 PTX12008 Fan tray
109 Fan Tray 1 REV 08 750-127041 BCCG8484 PTX12008 Fan tray
110 Fan Tray 2 REV 08 750-127041 BCCG8481 PTX12008 Fan tray
111 POWERSHELF 0 BUILTIN BUILTIN 5 Row PSM POWERSHELF
112
113 root@ptx12008> show interfaces terse et-0/0/0
114 error: device et-0/0/0 not found
115
116 root@ptx12008> show interfaces terse et-0/0/1

```

```
117 error: device et-0/0/1 not found
118
119 root@ptx12008> show interfaces terse et-0/0/2
120 error: device et-0/0/2 not found
121
122 root@ptx12008> show chassis power detail | match FPC
123   FPC 0 1179
```

By marking 3 ports as unused, Line card consumed power came down from 1234 to 1179 watts.

This behavior is reversible with “delete interface et-x/y/z unused” and it just takes a few seconds to restore the port in its original state, without any impact on the other ports of the system.

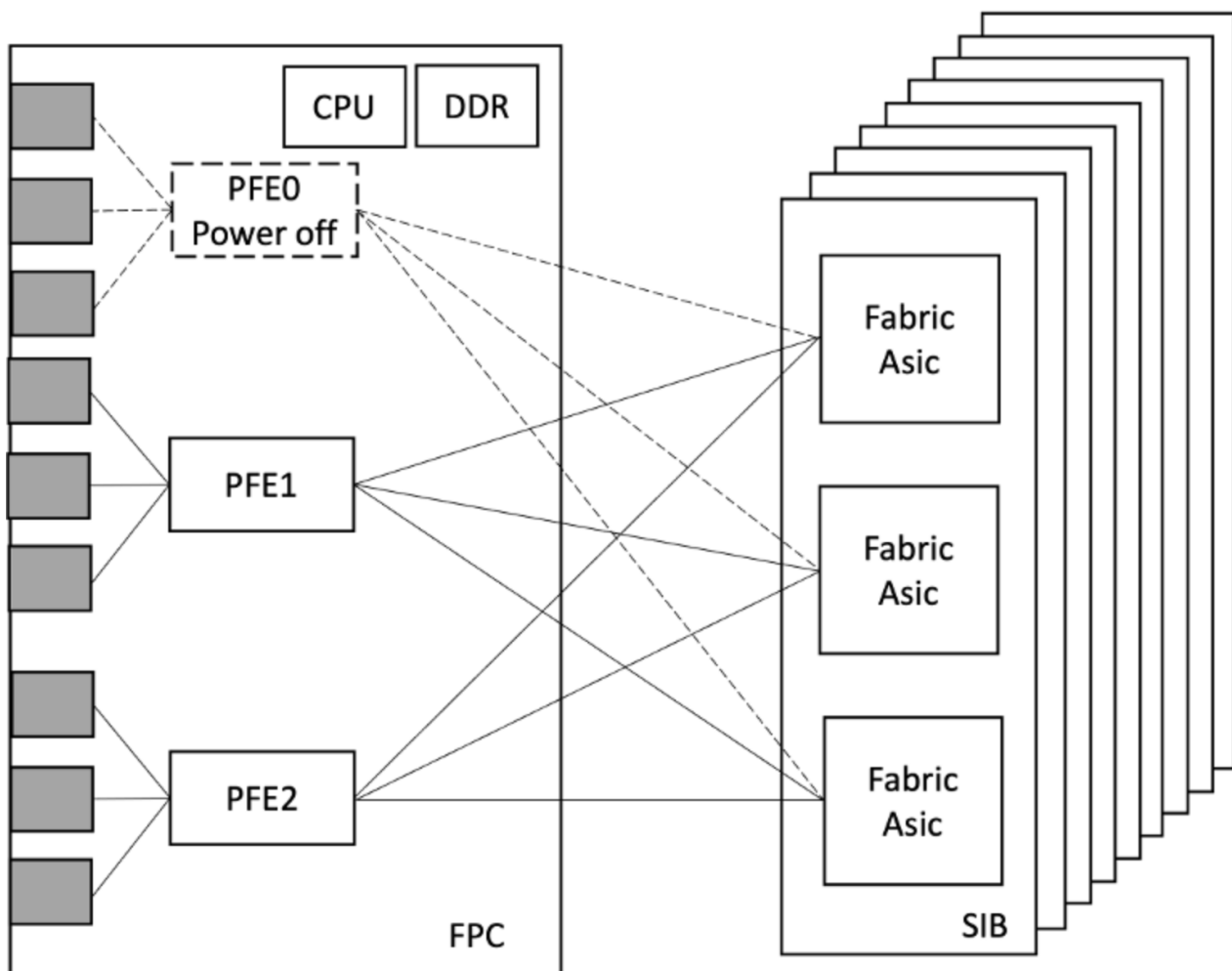
1.6 Selective PFE Power off

Each PFE (Packet Forwarding Engine ASIC) on an Line card can be powered off independently without affecting other PFEs on the same Line card or in the system. This is one of the most impactful power-saving features available to operators.

Use cases:

- Pay-As-You-Grow: Only the PFEs needed for current port count are powered on. Unused PFEs remain off until capacity is expanded.
- Off-peak scheduling: Power off idle PFEs during low-traffic window (e.g., nights/weekends). Use traffic engineering to keep traffic away from those PFEs before powering down.

PFE Power off help to save Line card power by turning off PFE completely. Also, it helps save power consumed by Fabric Cards by turning off Fabric serdes that map to the offlining PFE.



Here is an example that displays power saving with PFE Power off

```

1 root@ptx12008> show chassis fpc 0 pfe-instance all
2
3 FPC 0
4 PFE-Instance PFE PFE-State
5 0 0 ONLINE
6 0 1 ONLINE
7 1 2 ONLINE
8 1 3 ONLINE
9 2 4 ONLINE
10 2 5 ONLINE
11
12 root@ptx12008> show chassis power detail | match FPC
13 FPC 0 2620
14 FPC 1 2102
15 FPC 2 2723
16 FPC 3 1943
17 FPC 4 2061
18 FPC 5 1462
19 FPC 6 2043
20 FPC 7 2321
21
22 root@ptx12008> show chassis power detail | match SIB
23 SIB 0 609
24 SIB 1 609
25 SIB 2 620
26 SIB 3 613
27 SIB 4 627
28 SIB 5 622
29 SIB 6 613
30 SIB 7 614
31 SIB 8 613
32
33 root@ptx12008> request chassis fpc slot 0 pfe-instance 0 offline
34 Fru Block offline initiated!
35

```

```

36 root@ptx12008> show chassis fpc 0 pfe-instance all
37
38 FPC 0
39 PFE-Instance PFE PFE-State
40 0 0 Offlined by CLI
41 0 1 Offlined by CLI
42 1 2 ONLINE
43 1 3 ONLINE
44 2 4 ONLINE
45 2 5 ONLINE
46
47 root@ptx12008> show chassis power detail | match FPC
48 FPC 0 1859
49 FPC 1 2097
50 FPC 2 2724
51 FPC 3 1935
52 FPC 4 2064
53 FPC 5 1460
54 FPC 6 2042
55 FPC 7 2325
56
57 root@ptx12008>
58
59 root@ptx12008> show chassis power detail | match SIB
60 SIB 0 577
61 SIB 1 572
62 SIB 2 588
63 SIB 3 581
64 SIB 4 594
65 SIB 5 590
66 SIB 6 580
67 SIB 7 582
68 SIB 8 579

```

Above example clearly shows that PFE Power off not just bring down Line card power consumption significantly but also helps to save power consumed by Fabric cards by turning off Fabric card side serdes'es that are mating with PFE.

More Details on saving power using PFE power of can be found on following TechPost article

<https://juniper.github.io/techposts/saving-energy-on-ptx-with-pfe-power-off/article>

1.7 MACSEC Block Clock Gating

MACsec encryption/decryption Security block resides inside the forwarding ASIC. It operates on a per-Port-Group (PG) basis. Each forwarding ASIC has 9 Port Groups.

When no port in a Port Group has MACsec configured, the MACsec block for that Port Group is automatically disabled and doesn't consume power.

- Automatic. No configuration needed to save power.
- MACsec is activated only when explicitly configured on a port:

```
1 set security macsec interfaces et-X/Y/Z
```

- This is fully automatic and reversible: adding or removing MACsec configuration enables or disables the block transparently.

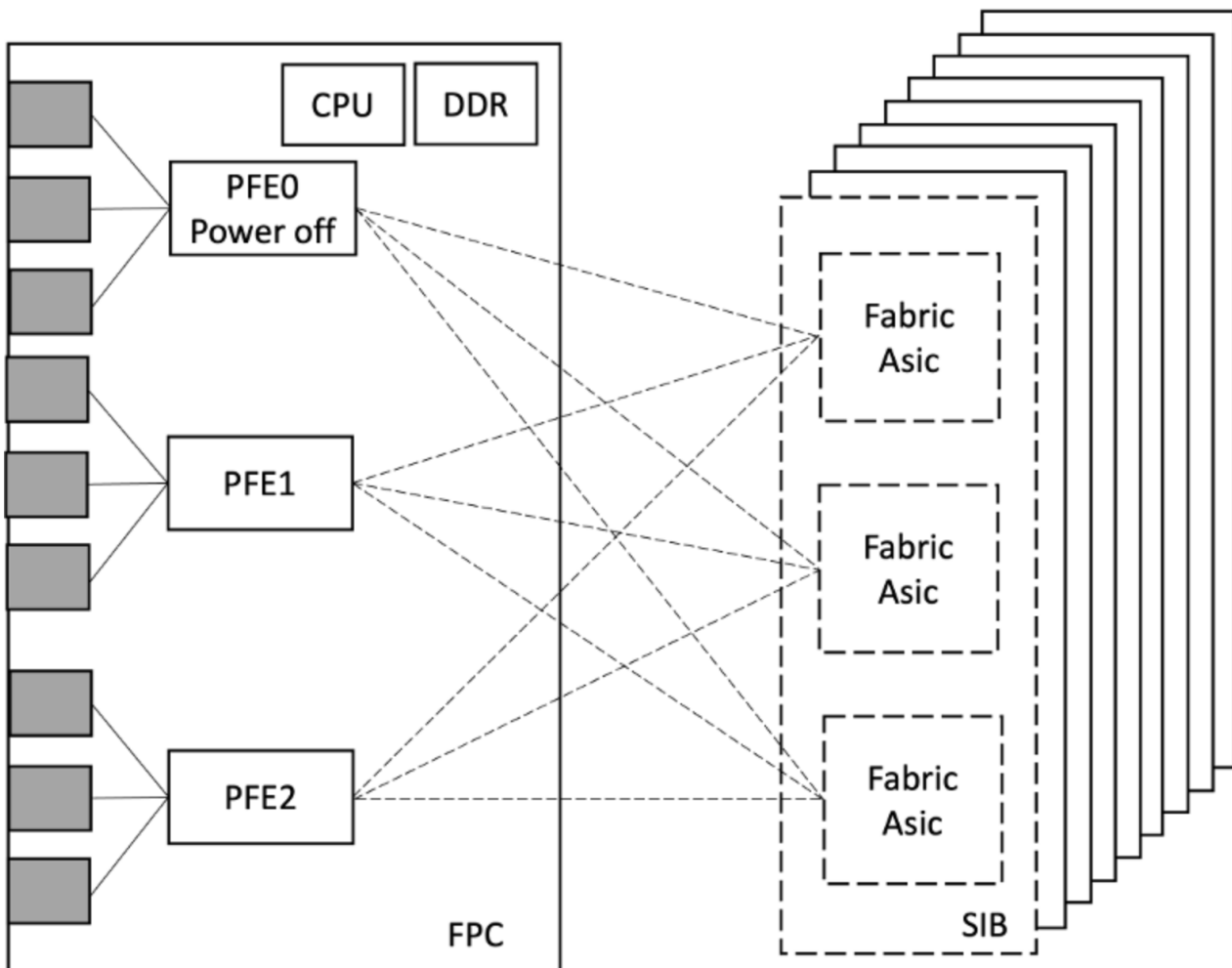
1.8 Fabric Card Power Off

The PTX12008 fabric card consists of multiple Switch Asics and the associated Fabric SerDes that form the full-mesh interconnect between all PFEs in the chassis.

An entire Fabric card can be taken offline. This is appropriate when:

- Reducing fabric power during sustained low-traffic periods when full fabric bandwidth is not required.
- Partial optic population per PFE: If fewer than the maximum number of optics are inserted per PFE, the fabric bandwidth requirement is proportionally lower and one or more SIBs can be kept offline. Example- If 8 out of 9 optics are inserted on all PFEs for PTX12008, the Fabric card does not need to carry full line-rate. 1 SIB can be kept offline without oversubscribing the remaining fabric.
- Lower bandwidth optics on WAN ports: If operators install lower-speed optics instead of the maximum supported rate, the aggregate bandwidth injected into the fabric is reduced, allowing some Fabric cards to be kept offline. Example - If 400G optics are installed on all PTX12008 Line card ports, the total fabric demand is halved. 4 Fabric cards can be offlined while the remaining SIBs carry the full traffic load without oversubscription.
- Scheduled Fabric card maintenance or replacement.

On Fabric Card offline, all Fabric Asics and serdes'es on Fabric card go to zero power. Along with this, Line card fabric serdes which are mating to the offlined fabric card, are also brought down and helps to save power on Line card as well.



Here is an example that displays power saving with Fabric Card Power off

```
1 root@ptx12008> show chassis sibs
2 Slot State Fabric links Errors
3 0 Online Active None
4 1 Online Active None
5 2 Online Active None
6 3 Online Active None
```

```

 7  4 Online Active None
 8  5 Online Active None
 9  6 Online Active None
10  7 Online Active None
11  8 Online Active None
12
13 root@ptx12008> show chassis power detail | match SIB
14   SIB 0 609
15   SIB 1 611
16   SIB 2 621
17   SIB 3 613
18   SIB 4 627
19   SIB 5 622
20   SIB 6 613
21   SIB 7 614
22   SIB 8 612
23
24 root@ptx12008> show chassis power detail | match FPC
25   FPC 0 2619
26   FPC 1 2099
27   FPC 2 2723
28   FPC 3 1942
29   FPC 4 2065
30   FPC 5 1460
31   FPC 6 2036
32   FPC 7 2326
33
34 root@ptx12008> request chassis sib slot 0 offline
35 Offline initiated, use "show chassis sibs" to verify
36
37 root@ptx12008> show chassis sibs
38 Slot State Fabric links Errors
39  0 Offline Unused None
40  1 Online Active None
41  2 Online Active None
42  3 Online Active None
43  4 Online Active None
44  5 Online Active None
45  6 Online Active None
46  7 Online Active None
47  8 Online Active None
48
49 root@ptx12008> show chassis power detail | match SIB
50   SIB 0 0
51   SIB 1 606
52   SIB 2 621
53   SIB 3 613
54   SIB 4 626
55   SIB 5 622
56   SIB 6 613
57   SIB 7 613
58   SIB 8 612
59
60 root@ptx12008> show chassis power detail | match FPC
61   FPC 0 2497
62   FPC 1 1965
63   FPC 2 2603
64   FPC 3 1814
65   FPC 4 1931
66   FPC 5 1371
67   FPC 6 1908
68   FPC 7 2186

```

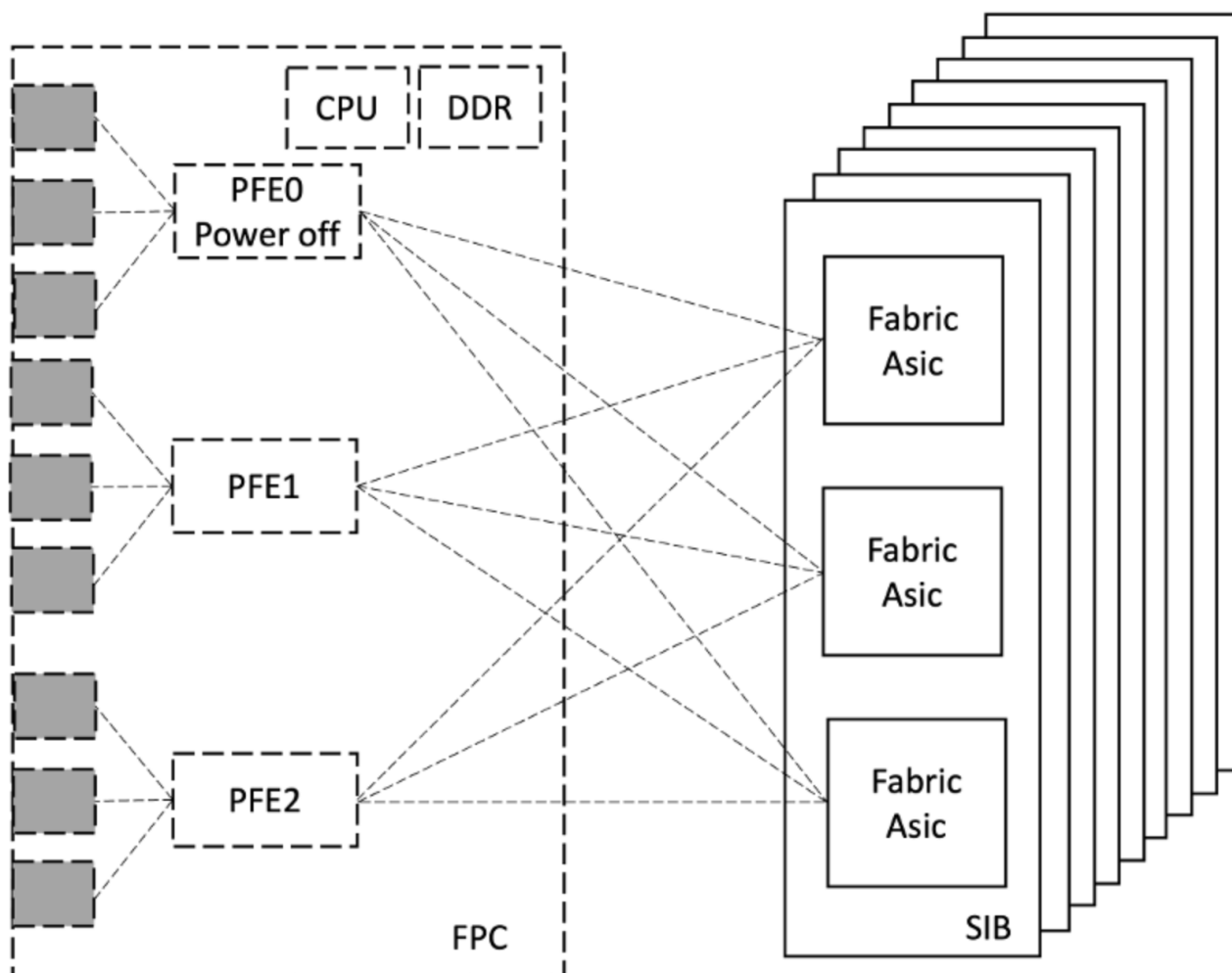
Above example clearly shows that Fabric card Power off not just bring down Fabric card power consumption to 0 but also helps to save power consumed by Line cards by turning off Line card side serdes'es that are mating with the offlined Fabric card.

1.9 Line Card Power Off

An entire Line card can be taken offline. This is appropriate when:

- Planning maintenance on a line card.
- Decommissioning capacity.

All PFEs, SerDes, PHYs and optics on the card go to zero power. Along with this, Fabric card serdes'es which are mating to the offlined Line card, are also brought down and it helps to save power on Fabric card as well.



Here is an example that displays power saving with Fabric Card Power off

```

1 root@ptx12008> show chassis fpc
2           Temp CPU Utilization (%) CPU Utilization (%) Memory Utilization (%)
3 Slot State (C) Total Interrupt 1min 5min 15min DRAM (MB) Heap Buffer
4 0 Online 48 5 0 5 5 65536 16 0
5 1 Online 39 4 0 4 4 65536 15 0
6 2 Online 45 4 0 4 4 65536 15 0
7 3 Online 38 4 0 4 4 65536 15 0
8 4 Online 40 4 0 4 4 65536 15 0
9 5 Online 36 4 0 4 4 65536 15 0
10 6 Online 36 4 0 4 4 65536 15 0
11 7 Online 42 4 0 4 4 65536 15 0
12
13 root@ptx12008> show chassis power detail | match SIB
14 SIB 0 609
15 SIB 1 604
16 SIB 2 621
17 SIB 3 613
18 SIB 4 626
19 SIB 5 622
20 SIB 6 613
21 SIB 7 613
22 SIB 8 612
23
24 root@ptx12008> show chassis power detail | match FPC
25 FPC 0 2618
26 FPC 1 2097
27 FPC 2 2727
28 FPC 3 1938
29 FPC 4 2055

```

```

30  FPC 5 1459
31  FPC 6 2034
32  FPC 7 2321
33
34  root@ptx12008> request chassis fpc slot 0 offline
35  Offline initiated, use "show chassis fpc" to verify
36
37  {master}
38  root@ptx12008> show chassis fpc
39                      Temp CPU Utilization (%) CPU Utilization (%) Memory Utilization (%)
40  Slot State (C) Total Interrupt 1min 5min 15min DRAM (MB) Heap Buffer
41  0 Offline ---Offlined by cli command---
42  1 Online 39 4 0 4 4 4 65536 15 0
43  2 Online 45 4 0 4 4 4 65536 15 0
44  3 Online 38 4 0 4 4 4 65536 15 0
45  4 Online 40 4 0 4 4 4 65536 15 0
46  5 Online 36 4 0 4 4 4 65536 15 0
47  6 Online 36 4 0 4 4 4 65536 15 0
48  7 Online 42 4 0 4 4 4 65536 15 0
49
50  root@ptx12008> show chassis power detail | match SIB
51  SIB 0 514
52  SIB 1 513
53  SIB 2 524
54  SIB 3 518
55  SIB 4 530
56  SIB 5 527
57  SIB 6 517
58  SIB 7 520
59  SIB 8 518
60
61  root@ptx12008> show chassis power detail | match FPC
62  FPC 0 0
63  FPC 1 2103
64  FPC 2 2728
65  FPC 3 1941
66  FPC 4 2060
67  FPC 5 1459
68  FPC 6 2042
69  FPC 7 2317

```

Above example clearly shows that Line card Power off not just bring down Line card power consumption to 0 but also helps to save power consumed by Fabric cards by turning off Fabric card fabric serdes'es that are mating with the offlined Line card.

1.10 Conclusion

The PTX12008 demonstrates that significant power savings can be achieved without compromising availability. The platform's layered approach, combining automatic mechanisms like Environmental Monitoring-driven fan control with operator-driven controls like Fabric Card/Line Card/PFE power-off, gives network operators fine-grained flexibility to match power draw to actual traffic demand. Cascading SerDes shutdown between Line cards and Fabric cards (and vice versa) ensures power savings propagate across the system rather than being isolated to a single component. Together, these capabilities make the PTX12008 a power-efficient platform well-suited for modern high-capacity networks where energy optimization is as critical requirement.

1.11 Useful links

- Power saving using power off: <https://juniper.github.io/techposts/saving-energy-on-ptx-with-pfe-power-off/article>
- TX10000 Power optimization: <https://juniper.github.io/techposts/ptx10000-power-optimization/article>

1.12 Glossary

- ASIC: Application Specific Integrated Circuit
- CPU: Central Processing Unit
- DDR: Double Data Rate
- EM: Environmental Monitoring
- FPC: Flexible PIC Concentrator
- IFD: Physical Interface
- PFE: Packet Forwarding Engine
- PG: Port Group
- PM: Power Management
- PSM: Power Supply Module
- SerDes: Serializer/Deserializer
- SIB: Switch Interface Board
- WAN: Wide Area Network



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